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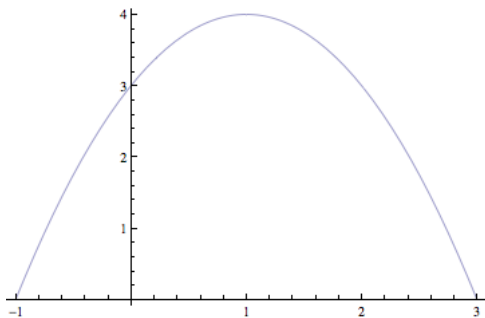
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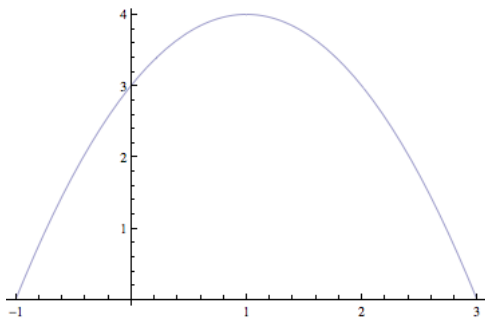
Example

Consider the function $f(x) = -x^2 + 2x + 3$



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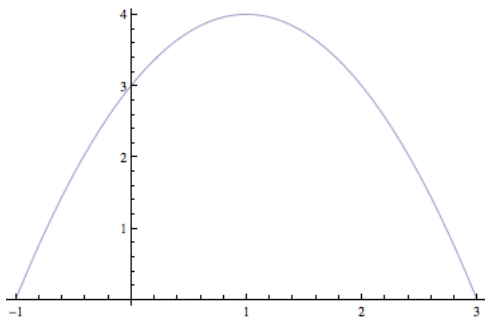
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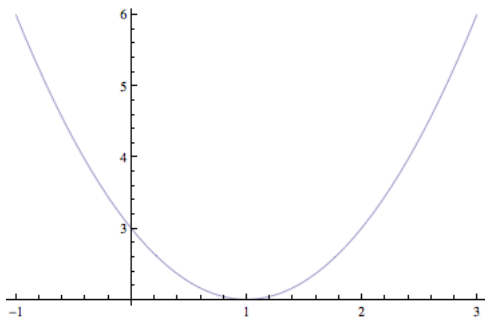
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- $f'(x) = -2x + 2$
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- $f'(x) = 0$ when $x = 1$
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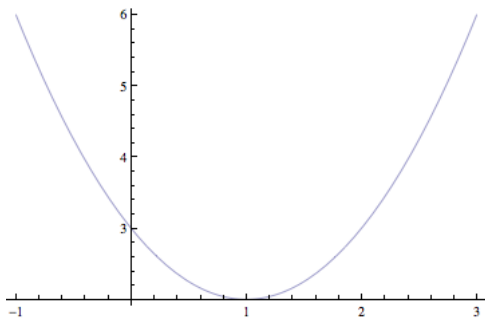
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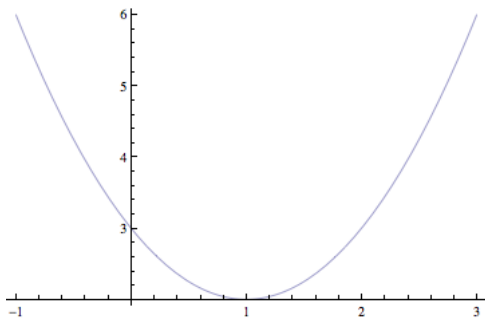
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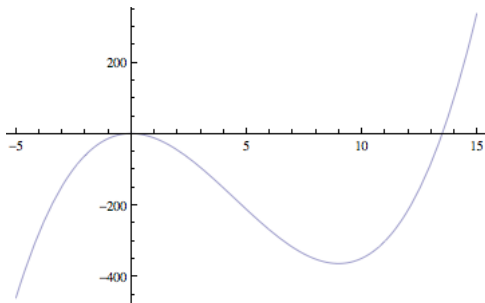
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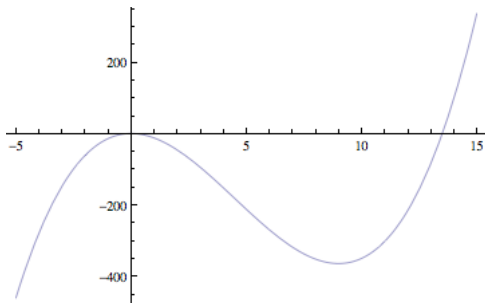
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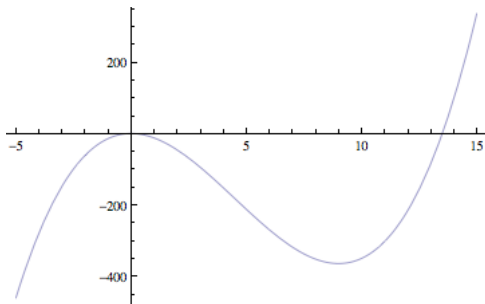
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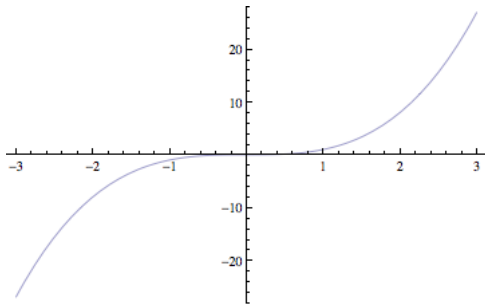
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- $f'(x) > 0$, when $x < 0$ or $x > 9$
- $f'(x) = 0$, when $x = 0$ or $x = 9$
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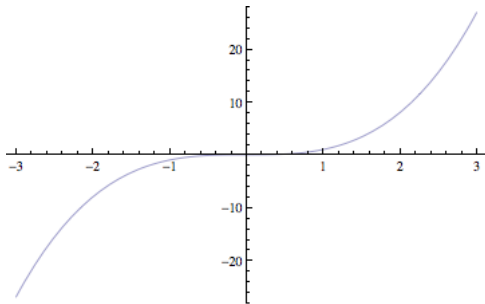
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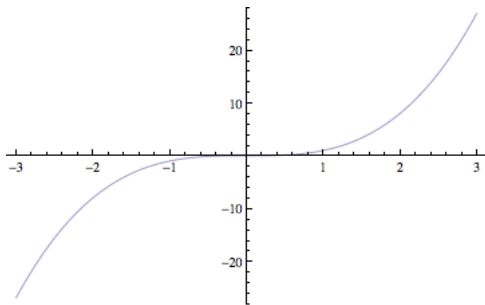
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- $f'(x) = 3x^2$
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Critical Points

- A function f has a *local maximum* at a point p if there exists $\delta > 0$ such that $f(q) \leq f(p)$ for all q such that $|p - q| < \delta$

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Thm. Let f be a function defined on the interval $[a, b]$. If f has a local maximum at a point $x \in (a, b)$ and if $f'(x)$ exists, then $f'(x) = 0$. The same is true for a local minima.

Rolle's Theorem

Thm: Suppose that $f : [a, b] \rightarrow \mathbb{R}^1$ is continuous on $[a, b]$, and C^1 on (a, b) . If $f(a) = f(b) = 0$ then there is a point $c \in (a, b)$ such that $f'(c) = 0$. Proof is on SB pg. 824.

Mean Value Theorem

Thm. Let $f : U \rightarrow \mathbb{R}^1$ be a C^1 function on a (connected) interval in U in $\mathbb{R}^1$. For any points $a, b \in U$, there is a point c between a and b so that: $f(b) - f(a) = f'(c)(b - a)$.

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Note that $g_0(a) = 0$ and $g_0(b) = 0$. By Rolle's theorem, there is a point $c \in (a, b)$ so that $g'_0(c) = 0$. Differentiating (2) with respect to x yields

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Plugging $x = c$ and $g'_0(c) = 0$ in this equation yields $f(b) - f(a) = f'(c)(b - a)$.

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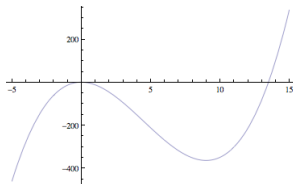
This theorem says that if you average 50 miles per hour on a trip from New York to Los Angeles, then at some point in the trip, you had to have been traveling exactly 50 miles per hour. For example if $f(x)$ is a function that assigns distance to time, x . Then the average miles per hour is given by $\frac{f(b)-f(a)}{b-a}$, which is equal to $f'(c)$, where c is some amount of time that is less than or equal to b and greater than or equal to a .

Global Maxima and Minima

Sometimes the local maxima is not equivalent to the global maxima. For example:

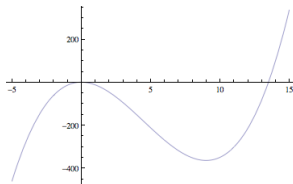
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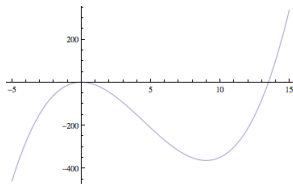
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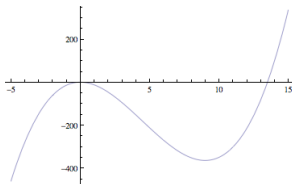
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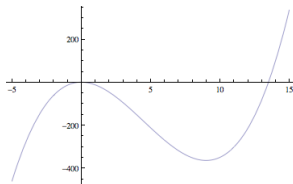
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 - Extrema that do not occur on the boundary are called **Interior Extrema**.

Formal definitions of Boundedness

Let A be a nonempty subset of $\mathbb{R}$. The set of the upper bounds of A , denoted $U(A)$ is defined as

$$U(A) = \{u \in \mathbb{R} \mid u \geq a \forall a \in A\}$$

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If $U(A)$ is nonempty, then A is said to be bounded above. If $L(A)$ is non-empty A is said to be bounded below

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Supremum and Infimum

- If $U(A)$ is nonempty, then the supremum of A , denoted $\sup A$ is defined to be the unique point $a^* \in U(A)$ such that $a^* \leq u \forall u \in U(A)$. If $U(A)$ is empty then A has no finite upper bound and, by convention, we set $\sup A = +\infty$.

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Higher Order Derivatives

- **C^1 functions:** $\frac{d}{dx}f(x)$ is also a function of x . If $\frac{d}{dx}f(x)$ is a continuous function, we call f *continuously differentiable*. Functions that are continuously differentiable are called C^1 functions.

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- **C^2 functions:** We can also differentiate $\frac{d}{dx}f(x)$, which is denoted as $\frac{d^2}{dx^2}$ or $f''(x)$ and is called the second order derivative. If the second order derivative is continuous, we say that f is twice continuously differentiable. This class of functions are called C^2 functions.

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- The k th order partial derivative is denoted $\frac{d^k}{dx^k}f(x)$ or $f^{(k)}(x)$. This class of function is called C^k functions.

Notation for Higher Order Derivatives

Second order derivatives can have the following notation $f''(x)$, $\frac{d^2 f(x)}{dx^2}$, or D_x^2 . This generalize for an n th order derivative: $f^{(n)}(x)$, $\frac{d^n f(x)}{dx^n}$, or D_x^n . A higher order partial derivative is defined as ∂_x^n .

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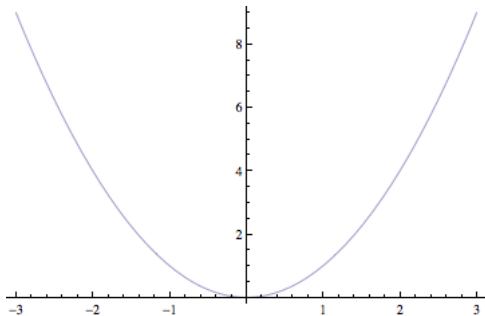
- $f'(x) = 3x^2 + 2$

- What is the second order derivative of this function?

- $f''(x) = 6x$

Concept of 2nd Order Derivative

Consider the function $f(x) = x^2$. The slope of this function at x is $f'(x) = 2x$. This slope is changing at a rate of $f''(x) = 2$. For example the slope at $f(1)$ is 2. The slope at $f(2) = 4$. This function is accelerating.



Concavity and Convexity

- **Concavity:** The function f is *concave* if $\forall \alpha \in [0, 1]$ and $x_1, x_2 \in \mathbb{R}$, $f(\alpha x_1 + (1 - \alpha)x_2) \geq \alpha f(x_1) + (1 - \alpha)f(x_2)$
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- These are different concepts than a convex set
- Concave curves have a negative second order derivative and
Convex curves have a positive second order derivative

Finding Extrema: First Derivative Test

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 - $f'(x^*) = 2x^* - 8 = 0 \implies x^* = 4$

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- If $f''(x^*) > 0$, then the point is a local minimum since the function is convex near x^* .

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- We can verify that the point x^* is an inflection point by graphing the function

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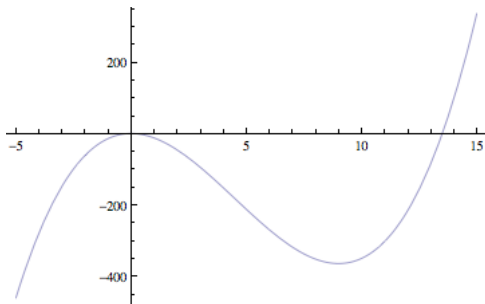
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- If the derivative is of odd order, then the point is an inflection point.
- If the derivative is of even order then point is an extrema. If the value of the non-zero derivative is negative, then the point is a maximum. If it is positive, then the point is a minimum.

Boundaries and Global Extrema

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- Consider the function $f(x) = x^3 - \frac{27}{2}x^2$ defined on the domain of $[-5, 15]$. What are the critical points? What is the global maximum?



Extreme Value Theorem

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- To find a global maximum and minimum
 - Check all local extrema
 - Check the boundary points
- For the function $f(x) = x^3 - \frac{27}{2}x^2$ defined on the interval of $[-5, 15]$, we know there is a local maximum at $x = 0$, which yields $f(0) = 0$. Then we only need to check the boundary points $f(-5) = -462.5$ and $f(15) = 337.5$

Taylor Polynomials

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By the definition of the derivative, $\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} = f'(a)$, we know $\frac{R(h;a)}{h} \rightarrow 0$ as $h \rightarrow 0$. So we know that $R(h; a)$ goes to zero and it goes to zero faster than h . *This means that the difference between the two sides of the first equation will be orders of magnitude smaller than h . We will show that with an infinitely differentiable function, Taylor polynomials can approximate the function arbitrarily well!!*

Taylor Polynomials Continued

Thm. 4 Let $f : U \rightarrow \mathbb{R}^1$ be a C^2 function defined on a (connected interval $U \in \mathbb{R}^1$). For any points a and $a + h$ in U , there exists a point c_2 between a and $a + h$ such that:

$$f(a + h) = f(a) + f'(a)h + \frac{1}{2}f''(c_2)h^2$$

Proof see SB pg. 828

This equation is just the one-term Taylor approximation for $f(a + h) = f(a) + f'(a)h + R(h; a)$, where the remainder satisfies: $R_1(h; a) = \frac{1}{2}f''(c_2)h^2$ for some c_2 between a and $a + h$. One can easily see that $\frac{1}{2} \frac{f''(c_2)h^2}{h} \rightarrow \frac{1}{2}f''(c_2) \cdot 0 = 0$, as $h \rightarrow 0$.

Two-Term Taylor Polynomials

To get a better approximation we can use the Taylor polynomial of f of order two:

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Then we can define the remainder as the difference in both sides:

$$R_2(h; a) \equiv f(a + h) - f(a) - f'(a)h - \frac{1}{2}f''(a)h^2$$

Two-Term Taylor Polynomials Continued

Thm Let $f : U \rightarrow R^1$ be a C^3 function defined on a (connected) interval U in R^1 . For any points a and $a + h$ in U , there exists a point c^* between a and $a + h$ such that

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Two-Term Taylor Polynomials Continued

Thm Let $f : U \rightarrow \mathbb{R}^1$ be a C^3 function defined on a (connected) interval U in $\mathbb{R}^1$. For any points a and $a + h$ in U , there exists a point c^* between a and $a + h$ such that

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This equation is the two-term Taylor approximation where the remainder satisfies $\frac{R_2(h;a)}{h^2} = \frac{1}{3!}f'''(c^*)h^3$. Note that $\frac{R_2(h;a)}{h^2} \rightarrow 0$ as $h \rightarrow 0$ and $R_2(h;a) \rightarrow 0$ faster than $R_1(h;a) \rightarrow 0$. A two-term Taylor approximation is more precise than a one-term Taylor approximation.

General Taylor Polynomial Formula

Following the logic in the previous slide we can define a k th order Taylor polynomial as

$$\begin{aligned} f(a+h) \approx P_k(a+h) &= f(a) + f'(a)h + \frac{1}{2!}f''(a)h^2 + \dots + \frac{1}{k!}f^{[k]}(a)h^k \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} h^n \end{aligned}$$

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$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

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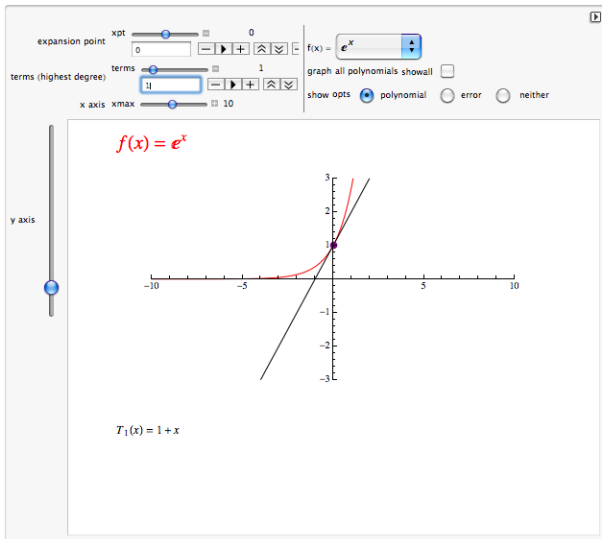
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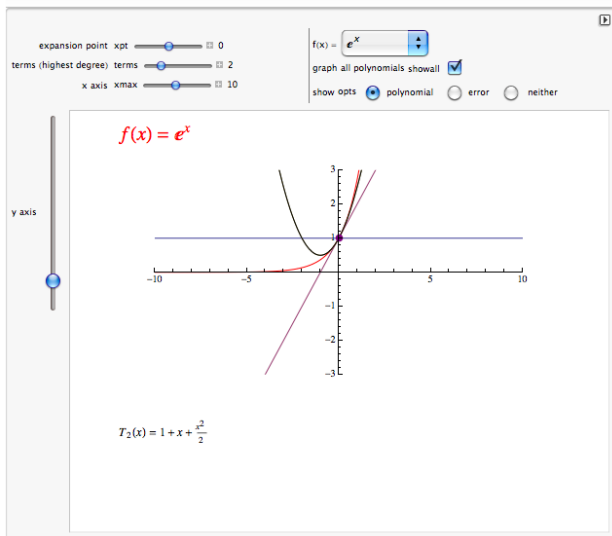
- $e^x = \sum_{n=0}^{\infty} \frac{h^n}{n!} = 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \dots$

- Where a , the expansion point is 0

Plotting Taylor Series



Plotting Taylor Series



Plotting Taylor Series

